

Live Forensics

F20FO/F21FO – Digital Forensics

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adapted from Emlyn Butterfield slides

Recap

- Incident Response Management
- Forensic Imaging
 - Forensic investigation toolkits
- Forensic Data Analysis
- Processing Evidence
- Hardware & Software Write Blocking

- What is live forensics
- Approach to live forensics
- Live forensics information sources
- Live forensics information collection

Learning Outcomes

- Describe what is live forensics
- Identify situations when to use live forensics
- Perform live forensics and data recovery

- **What is live forensics**
- Approach to live forensics
- Live forensics information sources
- Live forensics information collection

- Live forensics
 - Process by which digital evidence is collected from a live (running) computing device
- Differs from static/dead forensics which involves data collection from a disk image
- Allows for collection of volatile data
- Avoids system interruption (though risks modifying system)

Live Forensics: Need & Opportunity

Need for Live Forensics

- Large storage disks
 - Terabyte systems are big and common
 - Searching (or indexing) takes time
 - Mirroring takes time
- Minimal downtime (mission critical systems)
- Harder to seize systems (even with court order)

Live Forensics is also an **opportunity**

- Fast: Low-profile examination
- Data access: Longer data lifetime, and exclusive access to data that might only be in RAM
- Context: Provide context for static analysis

Data Lifetime

- Most current systems take no steps to overwrite sensitive data in memory
- Most applications that handle sensitive data weren't specifically designed to deal with sensitive data
 - Word processors:
 - Financial data, medical records, lists of passwords
- Password entered in web browser may be duplicated dozens of times
 - kernel buffers, window manager, application buffers, dynamic memory allocator
- While such data is often protected in data storage (e.g., encrypted on disk), it's not often protected in volatile memory

Live Forensics Goals

- What's happening now
- Who is doing what
- Obtain another piece of the puzzle
- Help focus static analysis (what parts of the disk)
- Get passwords, unencrypted data, etc.
- Information can be used to:
 - Reconstruct sessions (e.g., web, ftp, telnet, IM)
 - Find files (downloaded or accessed through network drives)
 - Find passwords
 - Identify remote machines

Constraints of Live Forensics

Constraints

- Access to system is required (even if time-limited)
- Must minimise changes and artifacts
 - analogy to physical data
 - must balance tool footprint to usefulness
- Need to minimise impact on system, particularly if it's "mission-critical"
 - Remember ACPO Principle 1!

Minimise Changes & Artifacts

- Small footprint (using trusted software)
- Try not to change anything
 - But everything changes, all the time
 - Minimise changes to evidence
 - Record all steps taken and artifacts
- Low profile, minimise detection
- Artifacts can be explained
 - analogy: detective's fingerprints on ransom note

Dangers of Live Forensics

- Most live forensics tools rely on the OS to provide some basic services
 - Reading of physical memory
 - Disk dumping
- But what if there's unexpected software on the machine that might impact live forensics
- Such as unexpected access controls
- Or malware, such as rootkits!

Dangers of Live Forensics - Rootkits

Two Categories of rootkits

- **User-Mode rootkits**
 - Shallow trickery
 - Modify system commands (e.g., ls, ps, du, df)
 - Change disk space statistics, list of running processes, etc.
- **Kernel-Mode rootkits**
 - Deep trickery
 - Modify OS to produce arbitrary results
 - Allows files, blocks of physical memory, etc. to be hidden even if trusted executables are used

Dangers of Live Forensics - Rootkits

- Disk drivers
- Hacking virtual memory system
- Replacement of shared libraries
- Affects even your “trusted” executables unless they’re statically linked!

- What is live forensics
- **Approach to live forensics**
- Live forensics information sources
- Live forensics information collection

Evidence Acquisition and Analysis

- First response/triage
- Looking for evidence, or for what computers, disks, directories, etc. may contain evidence
- Example
 - 30 servers, search warrant says image 1
 - search for quick hash scan to find sufficient cause to get a broader warrant
- Looking for context for static analysis

Require Access to System

- Before investigation
 - Use pre-installed agents
 - Requires prior access
 - Agents must not have become compromised
- During investigation
 - Need credentials for log-in
 - Must use known good binaries
 - Artifacts need to be accounted for

Scenario – Triage

- Limited time
- Want to answer:
 - Is there a problem?
 - What machine(s) are affected?
 - What disks need to be imaged?
 - What is running on the system?
- Focus on where the **problem is the worst** and the **evidence is the most abundant**

Scenario – Illegal Server

- Compare what's running on the machine and what ports are open to a network scan (e.g. nmap)
- What services does the world see visible on the computer?
- Some techniques are subtle (“port knocking”), but most are not
- Look at network traces to see who is talking to the computer (be careful of legal issues)

Scenario – Trojan Defence

“It’s not my fault, someone else is controlling my computer or hacked my account!”

- Support or refute the claim
 - Are there traces of (known) Trojans?
 - Are there unusual network connections?
 - What is running on the machine (that can be seen)?
 - Any indication that something is hidden?
 - Enough other evidence/activities to corroborate?
- Still relatively new in legal terms (1st used in court in UK in 2003)
- Still tricky (often Trojans/viruses are present)
- Plus, malware is often present that didn’t impact the case in a significant way

- What is live forensics
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- Live forensics information collection

Persistence of Memory

- Many systems retain at least some data after a warm reboot, reset, or even cold reboot
- Highly dependent on model and BIOS settings
- Potentially useful as a “last resort” for obtaining live forensics data, assuming computer model is known to have post-reboot persistent memory
- Research in the area of memory forensics

Information Available

- Images of entire disk
 - Live disk imaging
- Deleted files
 - Live file carving
- Unencrypted document fragments
- Encryption keys for whole-disk encryption schemes
- Copies of volatile-only malware (for disassembly and investigation)

Running Processes

- Windows

- Open files
- Open network connections
- Registry activity
- Open DLLs
- ...

- Unix

- Open files
- Open network connections
- Access to corresponding EXE, even if deleted
- Command line that invoked application
- Environment variables
- ...

Memory

- Process memory
 - Finer-grained than dumping entire RAM
 - Easier to make sense of virtual address space for a process than physical memory
 - More likely to find contiguous application structures
 - Can yield passwords, document fragments, unencrypted documents
- Kernel memory
 - Search for “hidden” processes
 - Evaluate health of kernel
- String searches
 - Most “brute force” technique

- Extract regular files
- Live disk imaging
 - Changing all the time
- Live file carving to retrieve deleted files
 - Need to be more creative than in traditional case
 - Trickier to attach sufficient high-speed storage

Virtual Machines

- Freeze VM
 - Copy/snapshot disks, memory, even screen
 - Resume execution
- Can also dump memory, disk images over local network connection between host and VM while VM runs
 - `nc -l -p 7000 -w 1 > dump.dd (host)`
 - `dd if=/dev/sda bs=512 | nc 192.168.1.50 7000 (within VM)`

- What is live forensics
- Approach to live forensics
- Live forensics information sources
- **Live forensics information collection**

- What's going on?
- Are things not right, processes or files hidden, or disk encryption in use?
- What's staying in memory (a lot) and process memory dumps?
- Dumping OS structures
 - determining which areas of swap space correspond to a particular process

sysinternals tools for Windows

- FileMon – shows filesystem activity in real time
- PSMon – watch process/thread creation in real time
- PsFile - shows files opened remotely
- PsKill - kill processes by name or process ID
- PsInfo - list information about a system
- PsList - list detailed information about processes
- PsLoggedOn - see who's logged on locally and via sharing
- PsLogList - dump event log records
- PsPasswd - changes account passwords
- PsService - view and control services
- PsSuspend - suspends processes
- Handle – shows which files are opened by which processes
- RegMon – see registry activity in real time
- ListDLLs – show loaded DLLs

Process Analysis Tools

- <https://vidstromlabs.com>
 - Pmdump** – dump process memory, given pid
 - ListDrivers** – list loaded kernel drivers
 - ListModules** – list EXEs / DLLs associated with a process
 - PromiscDetect** – determine if network adaptors are running in promiscuous mode
- There are many other tools

```
C:\WINDOWS\system32\cmd.exe

C:\>psinfo

PsInfo v1.73 - Local and remote system information viewer
Copyright (C) 2001-2005 Mark Russinovich
Sysinternals - www.sysinternals.com

System information for \\MONK:
Uptime:                0 days 0 hours 24 minutes 3 seconds
Kernel version:        Microsoft Windows XP, Multiprocessor Free
Product type:           Professional
Product version:        5.1
Service pack:           2
Kernel build number:    2600
Registered organization:
Registered owner:       golden
Install date:           10/5/2006, 1:42:47 PM
Activation status:      Error reading status
IE version:             6.0000
System root:            C:\WINDOWS
Processors:             2
Processor speed:        2.3 GHz
Processor type:          Intel(R) Core(TM)2 CPU           6600  @
Physical memory:        2814 MB
Video driver:            NVIDIA GeForce 7900 GS

C:\>
```


pslist

```

C:\WINDOWS\system32\cmd.exe
C:\>
C:\>
C:\>pslist

PsList 1.26 - Process Information Lister
Copyright (C) 1999-2004 Mark Russinovich
Sysinternals - www.sysinternals.com

Process information for MONK:

Name                Pid  Pri  Thd   Hnd   Priv   CPU Time   Elapsed Time
Idle                 0     0    2     0     0   233:33:08.671  0:00:00.000
System              4     8   86   694     0   0:04:39.640  0:00:00.000
smss                840    11    3    22    168   0:00:00.031  118:23:39.031
csrss               912    13   13   765   4660   0:02:26.171  118:23:20.328
winlogon            940    10   19   584   7460   0:00:02.953  118:23:18.406
services            984     9   15   336   2128   0:00:12.171  118:23:18.312
lsass               996     9   21   404   4392   0:00:47.796  118:23:18.296
svchost             1268    8   15   222   3260   0:00:00.390  118:23:17.203
svchost             1356    8   11   592   2128   0:00:02.656  118:23:11.812
svchost             1780    8   69  1797  26408   0:05:30.296  118:23:11.781
InCDsrv             1804    8   10   122   1084   0:00:00.046  118:23:11.765
svchost             1924    8    6   102   1472   0:00:00.578  118:23:11.718
svchost             308     8   14   212   1820   0:00:00.046  118:23:11.578
spoolsv             500     8   16   224   5012   0:00:04.093  118:23:11.359
CTSVCCDA            712     8    2    31    416   0:00:00.015  118:22:59.312
mdm                 784     8    3   107   1176   0:00:00.062  118:22:59.312
nsvsvc32            792     8    5   287   2276   0:00:01.031  118:22:59.296
SCWatch4           1408     8    4    36    936   0:00:00.031  118:22:53.234
svchost             1988     8    3   131   2604   0:00:00.140  118:22:53.203
TSIRCSRV            524     8    3   28    488   0:00:00.031  118:22:53.109
wdfmgr              552     8    4    67   1516   0:00:00.031  118:22:53.093
vmware-authd        1120     8    5   135   1584   0:08:07.906  118:22:53.062
vmount2             800     8    3    96   1100   0:00:00.031  118:22:52.953
vmnat               1064     8    3    59    676   0:00:00.046  118:22:52.640
vmnetdhcp           1708     8    2    35    536   0:00:00.031  118:22:52.578
alg                 3016     8    6   105   1180   0:00:00.015  118:22:50.671
TSIRCUSR            4004     8    3    24    420   0:00:00.015  118:22:08.921
explorer            4044     8   24   993  26440   0:04:22.015  118:22:08.859
CTHELPER            144     8    1    50   1792   0:00:00.109  118:22:08.328
CTXFIHLP            184     8    4   127  2248   0:00:00.140  118:22:08.312
nvraidservice       1884     8    3   110   1084   0:00:00.078  118:22:08.171
DMXLauncher         204     8    2    76    844   0:00:00.078  118:22:08.062
CTDVDDDET           240     8    1    56    836   0:00:00.046  118:22:08.015
VolPanel            244     8    4   150   5832   0:00:02.093  118:22:07.906
DLLML               272     8   10   172   5060   0:00:01.718  118:22:07.906
wmiprvse            1828     8    4   159   1996   0:00:00.062  118:22:07.687
CTXFISPI            1452     8    6   267   3520   0:00:00.765  118:22:07.578
DLACTRLW            1348     8    3    96   1072   0:00:00.109  118:22:07.265
issch               1520     8    1    22    360   0:00:00.015  118:22:06.875

```

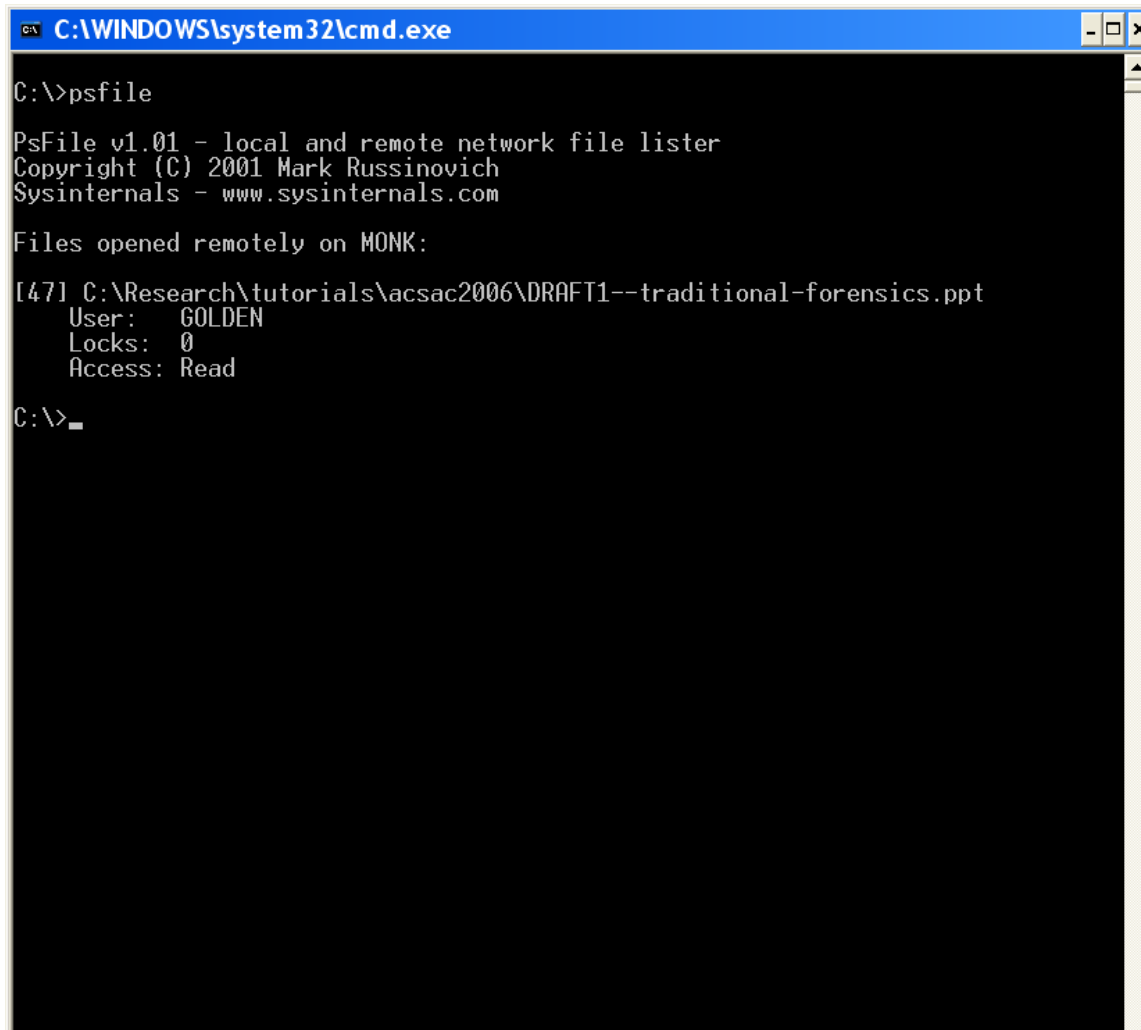
```
C:\WINDOWS\system32\cmd.exe
C:\>handle | tail -30
firefox.exe pid: 2652 MONK\golden
C: File (RW-) C:\Program Files\Mozilla Firefox
10: File (RW-) C:\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_659
5b64144ccf1df_6.0.2600.2982_x-ww_ac3f9c03
54: File (RW-) C:\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_659
5b64144ccf1df_6.0.2600.2982_x-ww_ac3f9c03
80: Section \BaseNamedObjects\CiceroSharedMemDefaultS-1-5-21-3900182140-
-3359249727-1237206558-1005
9C: Section \BaseNamedObjects\CTF.TimListCache.FMPDefaultS-1-5-21-39001
82140-3359249727-1237206558-1005SFM.DefaultS-1-5-21-3900182140-3359249727-123720
6558-1005
F4: File (---) C:\Documents and Settings\golden\Application Data\Mozilla\F
irefox\Profiles\vi97y8rj.default\parent.lock
1D4: File (RW-) C:\Program Files\Mozilla Firefox\chrome\browser.jar
1EC: File (RW-) C:\Program Files\Mozilla Firefox\chrome\toolkit.jar
204: File (RW-) C:\Program Files\Mozilla Firefox\chrome\classic.jar
208: File (RW-) C:\Documents and Settings\golden\Local Settings\Application
Data\Mozilla\Firefox\Profiles\vi97y8rj.default\XUL.mfl
210: File (RW-) C:\Documents and Settings\golden\Local Settings\Application
Data\Mozilla\Firefox\Profiles\vi97y8rj.default\Cache\CACHE_001_
214: File (RW-) C:\Documents and Settings\golden\Local Settings\Application
Data\Mozilla\Firefox\Profiles\vi97y8rj.default\Cache\CACHE_MAP_
218: File (RW-) C:\Documents and Settings\golden\Local Settings\Application
Data\Mozilla\Firefox\Profiles\vi97y8rj.default\Cache\CACHE_002_
21C: File (RW-) C:\Documents and Settings\golden\Local Settings\Application
Data\Mozilla\Firefox\Profiles\vi97y8rj.default\Cache\CACHE_003_
24C: Section \BaseNamedObjects\MSCTF.GCompartListSFM.DefaultS-1-5-21-390
0182140-3359249727-1237206558-1005
25C: Section \BaseNamedObjects\MSCTF.Shared.SFM.A0P
264: File (RW-) C:\Documents and Settings\golden\Application Data\Mozilla\F
irefox\Profiles\vi97y8rj.default\history.dat
2A0: File (RW-) C:\Documents and Settings\golden\Application Data\Mozilla\F
irefox\Profiles\vi97y8rj.default\formhistory.dat
2A4: File (RW-) C:\Documents and Settings\golden\Application Data\Mozilla\F
irefox\Profiles\vi97y8rj.default\cert8.db
2A8: File (RW-) C:\Documents and Settings\golden\Application Data\Mozilla\F
irefox\Profiles\vi97y8rj.default\key3.db
2B0: Section \BaseNamedObjects\tr_query_result_map_2
2E4: Section \BaseNamedObjects\ShimSharedMemory
2F0: File (RW-) C:\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_659
5b64144ccf1df_6.0.2600.2982_x-ww_ac3f9c03
-----
handle.exe pid: 2372 MONK\golden
BC: File (RW-) C:\
-----
tail.exe pid: 1524 MONK\golden
BC: File (RW-) C:\
C:\>
```

File Monitor - Sysinternals: www.sysinternals.com

File Edit Options Volumes Help

#	Time	Process	Request	Path	Result	Other
6306	11:40:16 AM	POWERP...	CLOSE	C:\Program Files\Trend Micro\Internet Security 12\TMAS_OE\TMAS_OEHook.dll	SUCCESS	
6307	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\Program Files\Trend Micro\Internet Security 12\TMAS_OE\TMAS_OEHook.dll	SUCCESS	Attributes: N
6308	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\Program Files\Trend Micro\Internet Security 12\TMAS_OE\TMAS_OEHook.dll	SUCCESS	Attributes: N
6309	11:40:16 AM	POWERP...	OPEN	C:\Program Files\Trend Micro\Internet Security 12\TMAS_OE\TMAS_OEHook.dll	SUCCESS	Options: Open Access: 001...
6310	11:40:16 AM	POWERP...	CLOSE	C:\Program Files\Trend Micro\Internet Security 12\TMAS_OE\TMAS_OEHook.dll	SUCCESS	
6311	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\DOCUME~1\golden\LOCALS~1\Temp\ppt4A72.tmp	SUCCESS	Attributes: A
6312	11:40:16 AM	POWERP...	OPEN	C:\DOCUME~1\golden\LOCALS~1\Temp\ppt4A72.tmp	SUCCESS	Options: Open Access: 000...
6313	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\DOCUME~1\golden\LOCALS~1\Temp\ppt4A72.tmp	SUCCESS	FileAttributeTagInformation
6314	11:40:16 AM	POWERP...	DELETE	C:\DOCUME~1\golden\LOCALS~1\Temp\ppt4A72.tmp	SUCCESS	
6315	11:40:16 AM	POWERP...	CLOSE	C:\DOCUME~1\golden\LOCALS~1\Temp\ppt4A72.tmp	SUCCESS	
6316	11:40:16 AM	POWERP...	CLOSE	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	
6317	11:40:16 AM	POWERP...	OPEN	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Options: Open Access: 001...
6318	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Length: 4488192
6319	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Length: 4488192
6320	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Length: 4488192
6321	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Attributes: A
6322	11:40:16 AM	POWERP...	LOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Excl Yes Offset: 214748353...
6323	11:40:16 AM	POWERP...	LOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Excl Yes Offset: 214748353...
6324	11:40:16 AM	POWERP...	UNLOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 2147483539 Length: ...
6325	11:40:16 AM	POWERP...	LOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Excl Yes Offset: 214748355...
6326	11:40:16 AM	POWERP...	UNLOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 2147483559 Length: ...
6327	11:40:16 AM	POWERP...	LOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Excl Yes Offset: 214748357...
6328	11:40:16 AM	POWERP...	UNLOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 2147483579 Length: ...
6329	11:40:16 AM	POWERP...	LOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Excl Yes Offset: 214748359...
6330	11:40:16 AM	POWERP...	UNLOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 2147483599 Length: ...
6331	11:40:16 AM	POWERP...	LOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Excl Yes Offset: 214748353...
6332	11:40:16 AM	POWERP...	LOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Excl Yes Offset: 214748355...
6333	11:40:16 AM	POWERP...	LOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Excl Yes Offset: 214748357...
6334	11:40:16 AM	POWERP...	LOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Excl Yes Offset: 214748359...
6335	11:40:16 AM	POWERP...	UNLOCK	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 2147483538 Length: 1
6336	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Attributes: A
6337	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Length: 4488192
6338	11:40:16 AM	POWERP...	SET INFORMATION	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Length: 4505600
6339	11:40:16 AM	POWERP...	READ	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 4489216 Length: 327...
6340	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Length: 4505600
6341	11:40:16 AM	POWERP...	SET INFORMATION	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Length: 4521984
6342	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Length: 4521984
6343	11:40:16 AM	POWERP...	SET INFORMATION	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Length: 4538368
6344	11:40:16 AM	POWERP...	READ	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 4521984 Length: 327...
6345	11:40:16 AM	POWERP...	QUERY INFORMATION	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Length: 4538368
6346	11:40:16 AM	POWERP...	SET INFORMATION	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Length: 4554752
6347	11:40:16 AM	POWERP...	FLUSH	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	
6348	11:40:16 AM	POWERP...	WRITE	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 0 Length: 4096
6349	11:40:16 AM	POWERP...	WRITE	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 376832 Length: 4096
6350	11:40:16 AM	POWERP...	WRITE	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 393216 Length: 4096
6351	11:40:16 AM	POWERP...	WRITE	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 4452352 Length: 4096
6352	11:40:16 AM	POWERP...	WRITE	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 4485120 Length: 614...
6353	11:40:16 AM	POWERP...	WRITE	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 0 Length: 4096
6354	11:40:16 AM	POWERP...	WRITE	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 393216 Length: 4096
6355	11:40:16 AM	POWERP...	WRITE	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 4452352 Length: 4096
6356	11:40:16 AM	POWERP...	WRITE	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 4485120 Length: 614...
6357	11:40:16 AM	POWERP...	FLUSH	C:\research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	
6358	11:40:16 AM	POWERP...	WRITE	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 376832 Length: 4096
6359	11:40:16 AM	POWERP...	WRITE	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 1056768 Length: 8192
6360	11:40:16 AM	POWERP...	WRITE	C:\Research\tutorials\acsac2006\live-forensics.ppt	SUCCESS	Offset: 1413120 Length: 4096

Psfile – remote file access



```
C:\WINDOWS\system32\cmd.exe

C:\>psfile

PsFile v1.01 - local and remote network file lister
Copyright (C) 2001 Mark Russinovich
Sysinternals - www.sysinternals.com

Files opened remotely on MONK:

[47] C:\Research\tutorials\acsac2006\DRAFT1--traditional-forensics.ppt
      User:    GOLDEN
      Locks:    0
      Access:  Read

C:\>_
```

promiscdetect

```
C:\WINDOWS\system32\cmd.exe

C:\>promiscdetect

PromiscDetect 1.0 - (c) 2002, Arne Vidstrom (arne.vidstrom@ntsecurity.nu)
                  - http://ntsecurity.nu/toolbox/promiscdetect/

Adapter name:
  - 1394 Net Adapter

Active filter for the adapter:

Adapter name:
  - Broadcom NetXtreme 57xx Gigabit Controller

Active filter for the adapter:
  - Directed (capture packets directed to this computer)
  - Multicast (capture multicast packets for groups the computer is a member of)
  - Broadcast (capture broadcast packets)

C:\>
```

Promiscdetect (Wireshark running)

```
C:\WINDOWS\system32\cmd.exe

C:\>promiscdetect

PromiscDetect 1.0 - (c) 2002, Arne Vidstrom (arne.vidstrom@ntsecurity.nu)
                  - http://ntsecurity.nu/toolbox/promiscdetect/

Adapter name:
- 1394 Net Adapter

Active filter for the adapter:

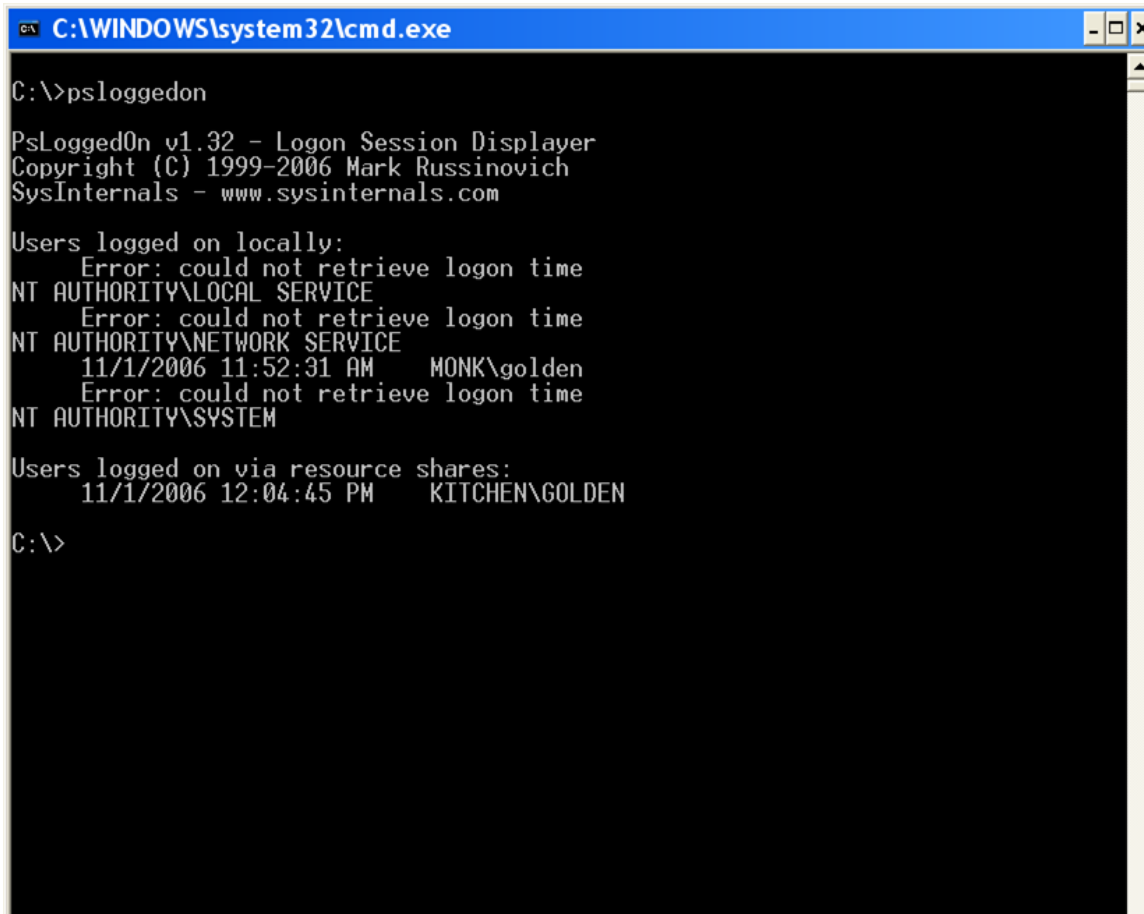
Adapter name:
- Broadcom NetXtreme 57xx Gigabit Controller

Active filter for the adapter:
- Directed (capture packets directed to this computer)
- Multicast (capture multicast packets for groups the computer is a member of)
- Broadcast (capture broadcast packets)
- Promiscuous (capture all packets on the network)

WARNING: Since this adapter is in promiscuous mode there could be a sniffer
        running on this computer!

C:\>
```

Psloggedon – logged on users



```
C:\WINDOWS\system32\cmd.exe

C:\>psloggedon

PsLoggedOn v1.32 - Logon Session Displayer
Copyright (C) 1999-2006 Mark Russinovich
SysInternals - www.sysinternals.com

Users logged on locally:
    Error: could not retrieve logon time
NT AUTHORITY\LOCAL SERVICE
    Error: could not retrieve logon time
NT AUTHORITY\NETWORK SERVICE
    11/1/2006 11:52:31 AM    MONK\golden
    Error: could not retrieve logon time
NT AUTHORITY\SYSTEM

Users logged on via resource shares:
    11/1/2006 12:04:45 PM    KITCHEN\GOLDEN

C:\>
```


- Memory dump

Plaintext

Netstat -a: all active connections

```

C:\WINDOWS\system32\cmd.exe

C:\>netstat -a

Active Connections

Proto Local Address           Foreign Address         State
TCP    monk:epmap               monk:0                  LISTENING
TCP    monk:microsoft-ds       monk:0                  LISTENING
TCP    monk:6881                monk:0                  LISTENING
TCP    monk:1027                monk:0                  LISTENING
TCP    monk:1042                localhost:1043          ESTABLISHED
TCP    monk:1043                localhost:1042          ESTABLISHED
TCP    monk:4664                monk:0                  LISTENING
TCP    monk:4845                localhost:4846          ESTABLISHED
TCP    monk:4846                localhost:4845          ESTABLISHED
TCP    monk:6999                monk:0                  LISTENING
TCP    monk:netbios-ssn        monk:0                  LISTENING
TCP    monk:microsoft-ds       192.168.2.210:3410     ESTABLISHED
TCP    monk:1040                by1msg2275509.phx.gbl:1863 ESTABLISHED
TCP    monk:1063                81.52.248.184:http     CLOSE_WAIT
TCP    monk:1064                81.52.248.184:http     CLOSE_WAIT
TCP    monk:3367                c-68-50-108-28.hsd1.md.comcast.net:37176 ESTABLISHED
ISHE
TCP    monk:3409                vili.demon.nl:13350    ESTABLISHED
TCP    monk:3579                C2ECZOTA_I11.chorzow.n4a.pl:50864 ESTABLISHED
TCP    monk:3656                pppoe-210.62.110.89-adsl.spbnit.ru:61288 ESTABLISHED
ISHE
TCP    monk:3684                dsl-202-72-137-238.wa.westnet.com.au:64730 ESTABLISHED
BLISHED
TCP    monk:4115                124-168-46-252.dyn.iinet.net.au:31998 ESTABLISHED
ED
TCP    monk:4210                host15-213-dynamic.3-87-r.retail.telecomitalia.it:6881 ESTABLISHED
t:6881 ESTABLISHED
TCP    monk:4440                86-125-26-227.oradea.rdsnet.ro:18663 TIME_WAIT
TCP    monk:4553                65.54.194.118:http     CLOSE_WAIT
TCP    monk:4707                cpe-68-206-116-246.stx.res.rr.com:7587 ESTABLISHED
HED
TCP    monk:4802                cpe-24-193-110-133.nyc.res.rr.com:65223 TIME_WAIT
IT
TCP    monk:4823                AC8DDCAF.ipt.aol.com:60191 TIME_WAIT
TCP    monk:4851                64.233.179.104:http    ESTABLISHED
TCP    monk:4852                static-fxfeeds.nslb.sj.mozilla.com:http ESTABLISHED
SHED
TCP    monk:4855                64.233.179.104:http    ESTABLISHED
TCP    monk:4861                64.233.179.104:http    ESTABLISHED
TCP    monk:4883                131.104.220.136:6882   TIME_WAIT
TCP    monk:4907                84-73-113-41.dclient.hispeed.ch:62215 TIME_WAIT
TCP    monk:4909                ip68-227-173-146.om.om.cox.net:6881 TIME_WAIT
TCP    monk:4910                70.Red-83-60-36.dynamicIP.rima-tde.net:6882 TIME_WAIT
  
```

Netstat -a -b: get exe too

```

C:\>netstat -a -b

Active Connections

Proto Local Address           Foreign Address         State       PID
TCP   monk:epmap              monk:0                  LISTENING   1360
c:\windows\system32\WS2_32.dll
C:\WINDOWS\system32\RPCRT4.dll
c:\windows\system32\rpcss.dll
C:\WINDOWS\system32\svchost.exe
-- unknown component(s) --
[svchost.exe]

TCP   monk:microsoft-ds       monk:0                  LISTENING   4
[System]

TCP   monk:6881               monk:0                  LISTENING   2824
[bittorrent.exe]

TCP   monk:1027               monk:0                  LISTENING   2992
[alg.exe]

TCP   monk:4664               monk:0                  LISTENING   2680
[GoogleDesktopIndex.exe]

TCP   monk:6999               monk:0                  LISTENING   2408
[tmproxy.exe]

TCP   monk:netbios-ssn        monk:0                  LISTENING   4
[System]

TCP   monk:netbios-ssn        monk:0                  LISTENING   4
[System]

TCP   monk:netbios-ssn        monk:0                  LISTENING   4
[System]

TCP   monk:1483               h181.64.255.206.cable.htsp.cablelynx.com:6881 SYN_SENT    2824
[bittorrent.exe]

TCP   monk:1490               ppp91-76-123-128.pppoe.mtu-net.ru:18768 SYN_SENT    2824
[bittorrent.exe]

TCP   monk:1497               ip-203-13.belltel.ph:6881 SYN_SENT    2824
[bittorrent.exe]

TCP   monk:1500               196.203.58.103:17367   SYN_SENT    2824
[bittorrent.exe]
  
```

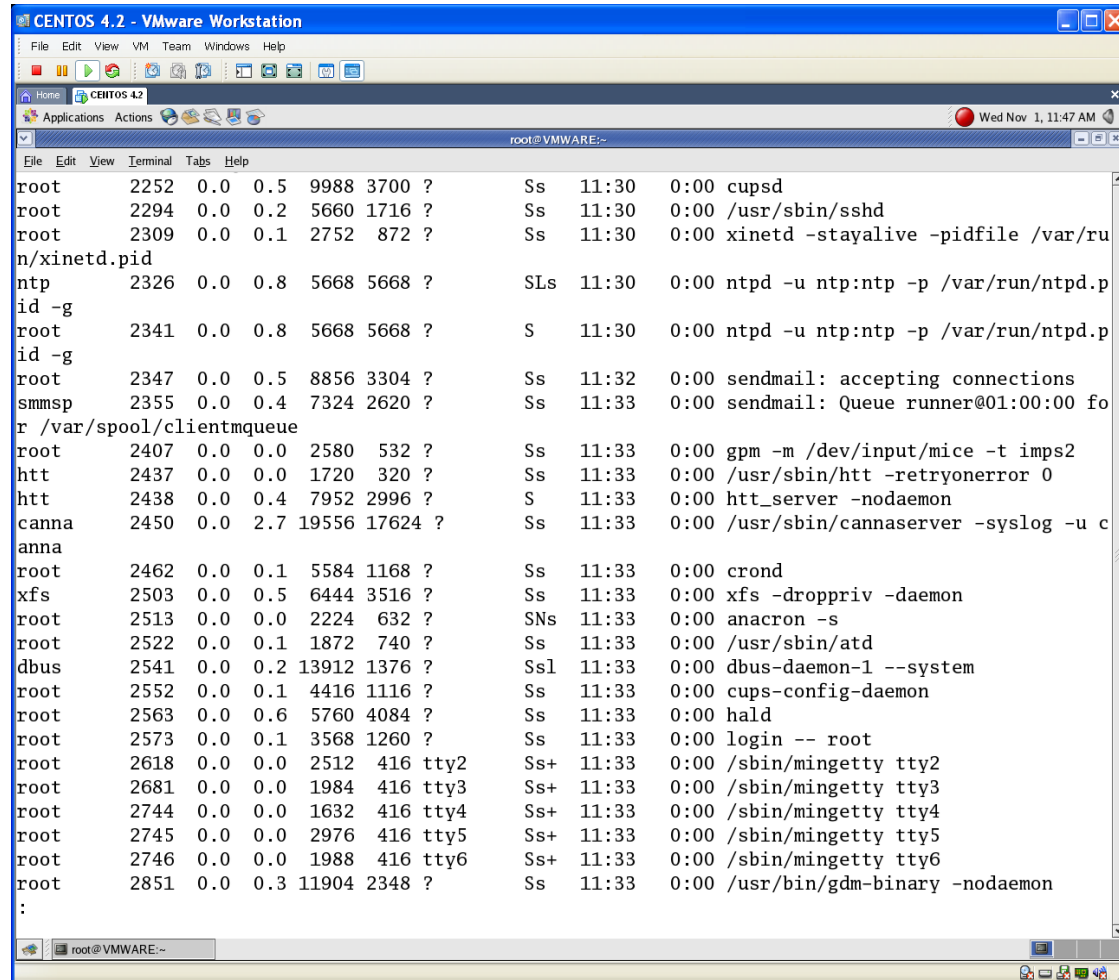
Linux Process Listing

Partial output of
`# ps aux | less`

a = show processes for all
users

u = display the process's
user/owner

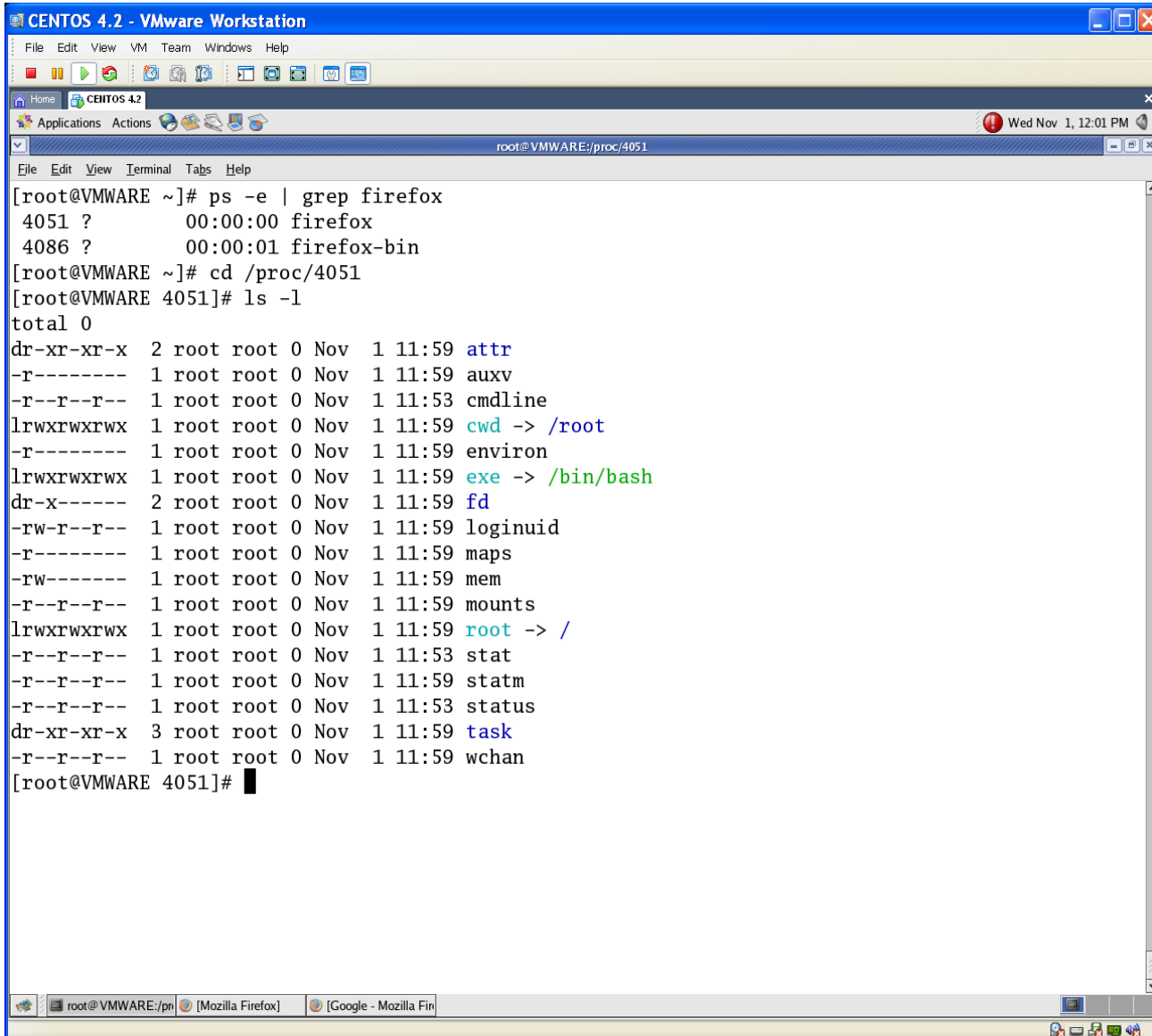
x = also show processes
not attached to a terminal



```

root@VMWARE:~$ ps aux
root      2252  0.0  0.5 9988 3700 ?        Ss   11:30   0:00 cupsd
root      2294  0.0  0.2 5660 1716 ?        Ss   11:30   0:00 /usr/sbin/sshd
root      2309  0.0  0.1 2752  872 ?        Ss   11:30   0:00 xinetd -stayalive -pidfile /var/run/xinetd.pid
ntp       2326  0.0  0.8 5668 5668 ?        SLs  11:30   0:00 ntpd -u ntp:ntp -p /var/run/ntpd.pid
root      2341  0.0  0.8 5668 5668 ?        S    11:30   0:00 ntpd -u ntp:ntp -p /var/run/ntpd.pid
root      2347  0.0  0.5 8856 3304 ?        Ss   11:32   0:00 sendmail: accepting connections
smmsp     2355  0.0  0.4 7324 2620 ?        Ss   11:33   0:00 sendmail: Queue runner@01:00:00 for /var/spool/clientmqueue
root      2407  0.0  0.0 2580  532 ?        Ss   11:33   0:00 gpm -m /dev/input/mice -t imps2
htt       2437  0.0  0.0 1720  320 ?        Ss   11:33   0:00 /usr/sbin/htt -retryonerror 0
htt       2438  0.0  0.4 7952 2996 ?        S    11:33   0:00 htt_server -nodaemon
canna     2450  0.0  2.7 19556 17624 ?    Ss   11:33   0:00 /usr/sbin/cannaserver -syslog -u c
anna
root      2462  0.0  0.1 5584 1168 ?        Ss   11:33   0:00 crond
xfs       2503  0.0  0.5 6444 3516 ?        Ss   11:33   0:00 xfs -droppriv -daemon
root      2513  0.0  0.0 2224  632 ?        SNs  11:33   0:00 anacron -s
root      2522  0.0  0.1 1872  740 ?        Ss   11:33   0:00 /usr/sbin/atd
dbus      2541  0.0  0.2 13912 1376 ?       Ssl  11:33   0:00 dbus-daemon-1 --system
root      2552  0.0  0.1 4416 1116 ?        Ss   11:33   0:00 cups-config-daemon
root      2563  0.0  0.6 5760 4084 ?        Ss   11:33   0:00 hald
root      2573  0.0  0.1 3568 1260 ?        Ss   11:33   0:00 login -- root
root      2618  0.0  0.0 2512  416 tty2    Ss+  11:33   0:00 /sbin/mingetty tty2
root      2681  0.0  0.0 1984  416 tty3    Ss+  11:33   0:00 /sbin/mingetty tty3
root      2744  0.0  0.0 1632  416 tty4    Ss+  11:33   0:00 /sbin/mingetty tty4
root      2745  0.0  0.0 2976  416 tty5    Ss+  11:33   0:00 /sbin/mingetty tty5
root      2746  0.0  0.0 1988  416 tty6    Ss+  11:33   0:00 /sbin/mingetty tty6
root      2851  0.0  0.3 11904 2348 ?        Ss   11:33   0:00 /usr/bin/gdm-binary -nodaemon
:
```

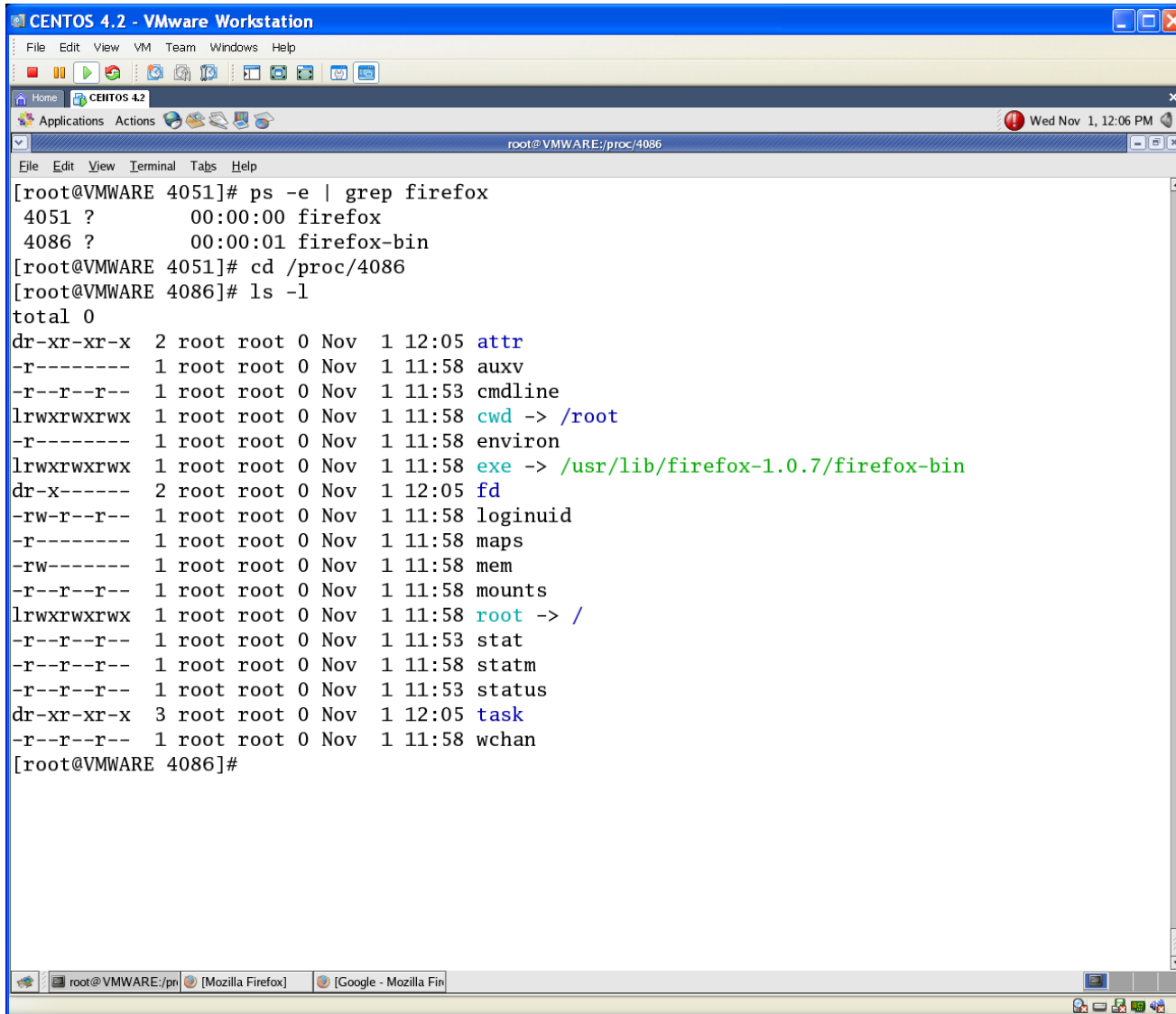
Linux: Detailed Process Info



```
CENTOS 4.2 - VMware Workstation
File Edit View VM Team Windows Help
Applications Actions
root@VMWARE:/proc/4051
File Edit View Terminal Tabs Help

[root@VMWARE ~]# ps -e | grep firefox
4051 ?          00:00:00 firefox
4086 ?          00:00:01 firefox-bin
[root@VMWARE ~]# cd /proc/4051
[root@VMWARE 4051]# ls -l
total 0
dr-xr-xr-x  2 root root 0 Nov  1 11:59 attr
-r-----  1 root root 0 Nov  1 11:59 auxv
-r--r--r--  1 root root 0 Nov  1 11:53 cmdline
lrwxrwxrwx  1 root root 0 Nov  1 11:59 cwd -> /root
-r-----  1 root root 0 Nov  1 11:59 environ
lrwxrwxrwx  1 root root 0 Nov  1 11:59 exe -> /bin/bash
dr-x-----  2 root root 0 Nov  1 11:59 fd
-rw-r--r--  1 root root 0 Nov  1 11:59 loginuid
-r-----  1 root root 0 Nov  1 11:59 maps
-rw-----  1 root root 0 Nov  1 11:59 mem
-r--r--r--  1 root root 0 Nov  1 11:59 mounts
lrwxrwxrwx  1 root root 0 Nov  1 11:59 root -> /
-r--r--r--  1 root root 0 Nov  1 11:53 stat
-r--r--r--  1 root root 0 Nov  1 11:59 statm
-r--r--r--  1 root root 0 Nov  1 11:53 status
dr-xr-xr-x  3 root root 0 Nov  1 11:59 task
-r--r--r--  1 root root 0 Nov  1 11:59 wchan
[root@VMWARE 4051]#
```

Linux: Detailed Process Info



```
CENTOS 4.2 - VMware Workstation
File Edit View VM Team Windows Help
Home
Applications Actions
root@VMWARE:/proc/4086
File Edit View Terminal Tabs Help
[root@VMWARE 4051]# ps -e | grep firefox
4051 ?      00:00:00 firefox
4086 ?      00:00:01 firefox-bin
[root@VMWARE 4051]# cd /proc/4086
[root@VMWARE 4086]# ls -l
total 0
dr-xr-xr-x  2 root root 0 Nov  1 12:05 attr
-r-----  1 root root 0 Nov  1 11:58 auxv
-r--r--r--  1 root root 0 Nov  1 11:53 cmdline
lrwxrwxrwx  1 root root 0 Nov  1 11:58 cwd -> /root
-r-----  1 root root 0 Nov  1 11:58 environ
lrwxrwxrwx  1 root root 0 Nov  1 11:58 exe -> /usr/lib/firefox-1.0.7/firefox-bin
dr-x-----  2 root root 0 Nov  1 12:05 fd
-rw-r--r--  1 root root 0 Nov  1 11:58 loginuid
-r-----  1 root root 0 Nov  1 11:58 maps
-rw-----  1 root root 0 Nov  1 11:58 mem
-r--r--r--  1 root root 0 Nov  1 11:58 mounts
lrwxrwxrwx  1 root root 0 Nov  1 11:58 root -> /
-r--r--r--  1 root root 0 Nov  1 11:53 stat
-r--r--r--  1 root root 0 Nov  1 11:58 statm
-r--r--r--  1 root root 0 Nov  1 11:53 status
dr-xr-xr-x  3 root root 0 Nov  1 12:05 task
-r--r--r--  1 root root 0 Nov  1 11:58 wchan
[root@VMWARE 4086]#
```

Software-based Acquisition of Physical Memory

- dd command
 - Under Linux:
 - /dev/mem (problematic now)
 - Physical memory
 - /proc/kcore
 - Kernel virtual memory
- Problem: must rely on OS to provide physical memory dump
- OS might be compromised
- After acquisition, interpretation is another issue!

Software-based Acquisition of Physical Memory

- Virtual machines (e.g., VMWare)
 - State of a virtual machine, including memory and disk, can be extracted
- Hibernation files
 - Snapshot of physical memory
 - Can hibernation process (e.g., writing of physical memory) be subverted?
 - Probably. Unaware of available tools to do it now

Problems with Memory Acquisition

- Software-based solutions for memory imaging typically require loading software
- Probably erases some evidence
- Requires at least limited trust of the OS
- Hardware solutions generally require pre-installed cards
- Potentially subject to nasty hardware poisoning tricks

Scenario: Viewing Memory for a Running Executable

- Let's look at an example of how we can view the memory of a running application
- Recall from Lecture Topic 3 that processes are used to manage the running of an application
- This includes managing the application's use of memory
- Memory for each process is structured into several segments, including
 - **Text:** Machine code of the program
 - **Stack:** Managing the function/method call structure during execution
 - **Heap:** Containing dynamically allocated memory

Scenario: Viewing Memory for a Running Executable

- Consider the following python program

```
$ python3
```

```
Python 3.12.3 (main, Jan 22, 2026, 20:57:42) [GCC 13.3.0]  
on linux
```

```
Type "help", "copyright", "credits" or "license" for more  
information.
```

```
>>> hello="Hello World!"
```

- The `hello` variable is allocated
- In the python environment, we can see the variable's memory location (in base 10)

```
>>> id(hello)
```

```
129592570691632
```

Scenario: Viewing Memory for a Running Executable

- Since each process has its own memory space, we need to find the process running python
- Particularly, we need the process ID (pid)
- Since we started an interactive python environment in our terminal, we can't run `ps` from the same terminal
- Below shows an example of active processes initiated within a terminal (`pts/0`), that includes a “testmem” application that is running in the background

```
$ ps
  PID TTY          TIME CMD
 4034 pts/0    00:00:00 bash
 4058 pts/0    00:00:00 testmem
 4065 pts/0    00:00:00 ps
```

- To find our process, we need to run `ps` from a different terminal window

Scenario: Viewing Memory for a Running Executable

- Thus, we can search for all processes (with the 'a' option) from another terminal and `grep` for our python process

```
$ ps aux | grep python3
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
mj	2963	0.0	0.0	4688	1284	pts/0	T	11:21	0:00	python3
mj	3107	0.0	0.0	9152	2392	pts/1	S+	11:28	0:00	grep python3

- Note that we ran the `ps` command on terminal `pts/1`, and our python script is running on terminal `pts/0`
- Now we have both the process identifier for our python program, and the starting memory location containing our variable
 - Starting memory address: 129592570691632
 - Process ID: 2963

Scenario: Viewing Memory for a Running Executable

- We can then use the `dd` command to help us to **read memory from a file** in the `/proc/` directory
- From Lecture Topic 3 `/proc/` is a *pseudo filesystem* with content stored in memory, not on disk

```
$ sudo dd bs=1 skip="129592570691632" count=1024 if="/proc/2963/mem" status=none |  
hexdump -C  
00000000  01 00 00 00 00 00 00 00  c0 62 a4 00 00 00 00  |.....b.....|  
00000010  0c 00 00 00 00 00 00 00  af 43 bb da 85 84 d6 24  |.....C.....$|  
00000020  64 00 00 00 00 00 00 00  48 65 6c 6c 6f 20 57 6f  |d.....Hello Wo|  
00000030  72 6c 64 21 00 00 00 00  00 00 00 00 00 00 00 00  |rld!.....|
```

- The above command converts the input file ("`if=`") corresponding to our process' memory store, starting at the memory location

Live Disk Imaging

- Image disks live using essentially the same tools as for preservation step in “static/dead” analysis
- Problem: Moving target, files changing
- On a relatively quiet system, image may be a “reasonable” representation

Win: `dd if=\\.\PhysicalDrive0 of=e:\pd0.dd`

Linux: `dd if=/dev/hdc of=/mnt/images/hdc.dd`

Live File Carving

- Similarly, can perform file carving against live block devices using standard tools
 - Scalpel, Foremost
- Beyond consistency problem, need sufficient available storage

Summary

- What is live forensics
- Approach to live forensics
- Live forensics information sources
- Live forensics information collection